

Three Exceptions Before Every Square

Sharp OEIS partition asymptotics, an all-orders coefficient calculus,
and phase-sensitive inverse transseries

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Abstract

Let $A(N)$ count the partitions of N whose parts do not exceed $\lfloor \sqrt{N} \rfloor$; this is OEIS A097356. A posted conjecture identifies $0.088154883798697116\dots$ as its lower asymptotic constant. We prove that this is exactly the sharp limit-inferior constant, while distinguishing that statement from a literal lower-bound inequality. The latter is eventually false at precisely three indices in every square block: the three integers immediately preceding the next square.

The proof is obtained from a uniform, all-orders expansion for $p_m(\alpha m^2 + \sigma m + \tau)$, where p_m counts partitions with parts at most m . An explicit finite Gaussian coefficient operator generates every correction. For A097356 the expansion has a nonvanishing phase factor $\rho^{\{\sqrt{N}\}}$ and polynomial corrections. We derive its cluster set, limiting distribution, summatory asymptotic, and inverse staircase. A Lambert- W_{-1} core suffices for the square subsequence, but the full sequence requires an additional piecewise-linear phase map: omitting it can cause an index error of order $\log Y$, whereas including it gives an $O(1)$ error. The shifted quadratic sequences A206227 and A206240 are covered by the same calculus.

The leading restricted-partition asymptotic is classical Szekeres theory, not a new discovery here. The endpoint classification and the detailed coefficient and inverse refinements are developed explicitly in this report. Analytic proofs, exact rational sign certificates, and high-precision checks are separated throughout. No complete exponentially improved transseries, effective last exceptional block, or machine-checked formalization is claimed.

Included artifacts. Editable $\text{\LaTeX}$, compiled PDF, two figures, reproducible Python code, exact rational certificates, and numerical tables.

[Write note, batch 73O1, 1 October 2026.] This is the collection edition of the report, written into ProveIt's research-report collection from manuscript 38 of batch 73O1 (provenance in appendix C). Every label carries the prefix `srp::`; the scripts now live in `code/` and the recorded outputs of the delivered `results/` directory in `data/`. Text added in the write is marked like this note; everything else is the delivered text. Two readings to keep in mind: the subtitle's word "transseries" is used loosely, since the expansions are phase-parametrized Poincaré expansions (the paragraph "Precise meaning of 'all orders'" in section 1 says so); and the discrete threshold proposition, proposition 11.2, is an instance of a theorem the repository already proves (see the note after it).

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1 The OEIS question and the scope of the result

1.1 Why this family is a worthwhile target

The restricted partition function

$$p_m(N) = [q^N]P_m(q), \quad P_m(q) = \prod_{j=1}^m (1 - q^j)^{-1}, \quad (1.1)$$

is a basic object in enumerative combinatorics. The cutoff $m \asymp \sqrt{N}$ is a natural two-parameter regime, not an artificially selected sparse sequence. In particular,

$$A(N) = p_{\lfloor \sqrt{N} \rfloor}(N) \quad (N \geq 1), \quad A(0) = 1 \quad (1.2)$$

is A097356. Its mild block oscillation comes from a deterministic integer cutoff; it is not erratic behavior of unknown origin.

The entry, inspected on October 1, 2026, credits Václav Kotešovec with a January 8, 2024 formula giving the square-subsequence constants and a conjectured lower constant for the complete sequence [1]. The wording makes two mathematically different readings possible:

$$\liminf_{N \rightarrow \infty} NA(N)d^{-\sqrt{N}} = C_-, \quad (1.3)$$

$$A(N) \geq C_- \frac{d^{\sqrt{N}}}{N} \quad \text{for every sufficiently large } N. \quad (1.4)$$

This report proves (1.3) and disproves (1.4), with an exact eventual classification of its violations. An asymptotic envelope need not be an eventual inequality; that distinction is the central mathematical point.

Three related entries supply useful comparisons:

Entry	Definition	Role here
A206226	$p_m(m^2)$	square subsequence
A206227	$p_m(m^2 + m)$	positive linear shift
A206240	$p_m(m^2 - m)$	negative linear shift

Their leading constants were already recorded in 2014 [2, 3, 4]. Notice that $p_m(m^2 - m)$ is *not* $A(m^2 - m)$: the latter generally has cutoff $m - 1$. Keeping this indexing distinction prevents a misleading identification of the lower-envelope constant.

1.2 What is classical, what is proved here, and what is not claimed

Szekeres' restricted-partition asymptotic is developed in Canfield's recursion-based proof and in Romik's probabilistic proof [6, 7]. Our leading term is a specialization of that classical theory. OEIS A206226 also already records the family $p_m(jm^2) \sim c(j)d(j)^m/m^2$ and a dilogarithmic equation for $d(j)$ [2]. We therefore do not count recovery of that leading term, by itself, as a new theorem of independent priority.

The specific results developed here are a uniform bounded-offset all-orders formula with an executable coefficient operator, its complete floor-phase conversion, the three-exception endpoint theorem, and phase-sensitive inverse formulas. The summatory law and the limiting distribution are additional consequences. The consulted sources do not supply these particular refinements in the form proved below; this is not an exhaustive priority certification of the entire literature.

The work follows the organization of ProveIt's transseries program: separate the dominant core, unit-series coefficients, remainder transport, and integer staircase. The inspected repository checkpoint

is 1f1981f682b2878bde51a6ad40c22777f362fc05. Its transseries README identifies relevant coefficient, Lambert-core, remainder-transport, and staircase modules [10]. This report supplies conventional mathematical proofs and executable checks, not a new Lean development. None of the analytic estimates below is assumed to follow automatically from that repository inventory.

Precise meaning of “all orders.” For every fixed K , the stated approximation has a proved remainder of the next indicated order. No convergence of the infinite correction series is asserted. The floor-dependent expansion is a phase-parametrized Poincaré expansion, not an ordinary power–logarithmic transseries of a smooth germ. Noncentral arcs are smaller than every algebraic order relative to the leading saddle, but their separate exponential sectors and Stokes data are not determined here.

2 Main results at a glance

Let $u > 0$ be the unique solution of

$$\int_0^1 \frac{x}{e^{ux} - 1} dx = 1. \quad (2.1)$$

Set

$$\rho = 1 - e^{-u}, \quad H = 2u - \log \rho, \quad d = e^H, \quad C = \frac{u}{2\pi\sqrt{2 - 3e^{-u}}}, \quad C_- = C\rho. \quad (2.2)$$

The square root in C is positive by the variance identity proved in section 3. Numerical values, included for orientation rather than as definitions, are

$$\begin{aligned} u &= 0.814651136747611096956850553134\dots, \\ H &= 2.214122138515842343608904165265\dots, \\ d &= 9.153370192454122461948530292401\dots, \\ C &= 0.158208720267250414976631099924\dots, \\ \rho &= 0.557206225104302225757470174656\dots, \\ C_- &= 0.088154883798697116516927278293\dots \end{aligned}$$

The value of d agrees with A258268 [5].

Theorem 2.1 (Phase expansion and sharp envelopes). *Put $t = \sqrt{N}$ and $\theta = \{\sqrt{N}\}$. For every fixed $K \geq 0$ there are explicit polynomials P_k , with $P_0 = 1$ and $\deg P_k \leq 2k$, such that*

$$A(N) = \frac{Ce^{Ht}}{t^2} \rho^\theta \left(\sum_{k=0}^K \frac{P_k(\theta)}{t^k} + O(t^{-K-1}) \right), \quad (2.3)$$

uniformly for all integer $N \rightarrow \infty$. In particular, for $R(N) = NA(N)e^{-H\sqrt{N}}$,

$$\liminf R(N) = C_-, \quad \limsup R(N) = C, \quad (2.4)$$

and the full set of subsequential limits is $[C_-, C]$.

Theorem 2.2 (Exactly three lower-envelope violations). *For every sufficiently large integer m , and every integer $m^2 \leq N < (m+1)^2$,*

$$A(N) < C_- \frac{d^{\sqrt{N}}}{N} \iff N \in \{(m+1)^2 - 3, (m+1)^2 - 2, (m+1)^2 - 1\}. \quad (2.5)$$

Furthermore $A(N) < Cd^{\sqrt{N}}/N$ for all sufficiently large N . The number of violations of the lower inequality up to X is $3\sqrt{X} + O(1)$.

The phrase “sufficiently large” is essential. The proof gives existence of a cutoff, not a certified smallest cutoff. For example, the numerical checks in section 13 show that the third-last point is still above the lower envelope in the block $m = 64$, and below it in the block $m = 128$.

Theorem 2.3 (Global phase-sensitive inverse). *Let $Q(Y) = \min\{N \geq 0 : A(N) \geq Y\}$ and define the large-branch Lambert core*

$$X(Y) = -\frac{2}{H} W_{-1} \left(-\frac{H}{2} \sqrt{\frac{C}{Y}} \right), \quad \gamma = \frac{2u}{H} = 0.735868290711 \dots \quad (2.6)$$

For $x = k + \eta$, $k = \lfloor x \rfloor$ and $0 \leq \eta < 1$, define

$$T(x) = k + \min \left(\frac{\eta}{\gamma}, 1 \right). \quad (2.7)$$

Then uniformly as $Y \rightarrow \infty$,

$$\sqrt{Q(Y)} = T(X(Y)) + O(X(Y)^{-1}), \quad Q(Y) = T(X(Y))^2 + O(1). \quad (2.8)$$

Replacing $T(X)$ by X can instead incur an index error of order $X \asymp \log Y$. Away from the phase transition points, the inverse has explicit all-orders interior expansions and exact square plateaus.

The rest of the report proves these statements, explains how to generate their coefficients, and records reproducible checks.

3 The saddle and the classical leading term

For the more general regime $N = \alpha m^2 + \sigma m + \tau$, define

$$F(z) = -\int_0^1 \log(1 - e^{-zx}) dx = \frac{\zeta(2) - \text{Li}_2(e^{-z})}{z}, \quad (3.1)$$

$$J(u) = \int_0^1 \frac{x}{e^{ux} - 1} dx = -F'(u), \quad (3.2)$$

$$\ell(z) = \frac{1}{2} \log \frac{z}{1 - e^{-z}}. \quad (3.3)$$

For real positive z these are unambiguous real functions. Analytic continuations used below are restricted to a small complex neighborhood of a positive saddle, with the branches continued from positive reals.

Lemma 3.1 (Existence, positivity, and useful identities). *For every $\alpha > 0$ there is a unique $u = u(\alpha) > 0$ satisfying $J(u) = \alpha$. It depends real-analytically on α . Define*

$$V = F''(u), \quad H = \alpha u + F(u), \quad C = \frac{e^{\ell(u)}}{2\pi\sqrt{V}}, \quad \rho = 1 - e^{-u}. \quad (3.4)$$

Then $V > 0$, and

$$F(u) = \alpha u - \log \rho, \quad H = 2\alpha u - \log \rho, \quad (3.5)$$

$$uV = 2\alpha - \frac{1}{e^u - 1}, \quad C = \frac{u}{2\pi\sqrt{2\alpha - (2\alpha + 1)e^{-u}}}. \quad (3.6)$$

Proof. Differentiation under the integral gives

$$J'(u) = -\int_0^1 \frac{x^2 e^{ux}}{(e^{ux} - 1)^2} dx < 0, \quad V = -J'(u) > 0.$$

The integral satisfies $J(u) \sim 1/u$ at zero, and $J(u) \leq u^{-2} \int_0^\infty y/(e^y - 1) dy \rightarrow 0$ at infinity. Existence and uniqueness follow by continuity, and analytic dependence follows from the implicit function theorem because $J'(u) \neq 0$.

Integration by parts in F gives $F(u) = -\log(1 - e^{-u}) + uJ(u)$; the boundary term at $x = 0$ is zero since $x \log x \rightarrow 0$. Write $J(u) = u^{-2} \int_0^u y/(e^y - 1) dy$ and differentiate to obtain $J'(u) = -2J(u)/u + 1/[u(e^u - 1)]$. This proves the variance identity. Substitution into $e^{\ell(u)}/(2\pi\sqrt{V})$ gives the displayed expression for C . Positivity of its radicand is therefore a consequence of $V > 0$, not a numerical assumption. $\square$

Comparison with the existing dilogarithmic constant. If $r = e^u$, differentiation and the value at $u = 0$ show

$$\text{Li}_2(1 - e^u) = -\frac{u^2}{2} - \int_0^u \frac{x}{e^x - 1} dx = -(\alpha + \frac{1}{2})u^2. \quad (3.7)$$

Consequently r solves $\text{Li}_2(1 - r) + (\alpha + 1/2) \log^2 r = 0$, and

$$e^H = \frac{r^{2\alpha+1}}{r - 1}. \quad (3.8)$$

This is the same parametrization recorded in A206226 [2]. In Szekeres' usual variable $t = m/\sqrt{N}$, the leading case has $t = \alpha^{-1/2}$; the relation $t^2 = u^2/\int_0^u x/(e^x - 1) dx$ in Canfield's formulation is exactly $J(u) = \alpha$ [6]. These identities explicitly reconcile our normalization with prior theory.

4 Euler–Maclaurin expansion of the finite partition product

The apparent singularity at the origin must be removed before applying Euler–Maclaurin. Put

$$h(z) = \log \frac{z}{1 - e^{-z}}, \quad h(0) = 0. \quad (4.1)$$

This function is analytic near zero, and along a small neighborhood of any compact positive-real segment.

Proposition 4.1 (All-orders logarithmic product). *For every $R \geq 0$, uniformly in a sufficiently small fixed complex neighborhood of a positive saddle,*

$$\log P_m(e^{-z/m}) = mF(z) - \frac{1}{2} \log(2\pi m) + \ell(z) + \sum_{r=1}^R \frac{E_{2r-1}(z)}{m^{2r-1}} + O(m^{-2R-1}), \quad (4.2)$$

$$E_{2r-1}(z) = -\frac{B_{2r}}{(2r)!} z^{2r-1} \text{Li}_{2-2r}(e^{-z}) - \mathbf{1}_{r=1} \frac{z}{24}. \quad (4.3)$$

The same estimate holds after any fixed number of local z -derivatives, with a possibly smaller neighborhood. The first two terms are

$$E_1(z) = -\frac{z}{24} \coth \frac{z}{2}, \quad E_3(z) = \frac{z^3}{720} \text{Li}_{-2}(e^{-z}). \quad (4.4)$$

Proof. Separate the logarithmic singularity exactly:

$$\log P_m(e^{-z/m}) = -m \log(z/m) - \log(m!) + \sum_{j=1}^m h(zj/m). \quad (4.5)$$

Apply Euler–Maclaurin to the last sum. Its integral term is $m \int_0^1 h(zx) dx$, its endpoint term is $h(z)/2$, and its $m^{-(2r-1)}$ term is

$$\frac{B_{2r}}{(2r)!} z^{2r-1} (h^{(2r-1)}(z) - h^{(2r-1)}(0)).$$

Stirling's expansion of $\log(m!)$ contributes $-B_{2r}/[2r(2r-1)]$ at the same order. Also $1 - \log z + \int_0^1 h(zx) dx = F(z)$.

For odd $p = 2r - 1$,

$$h^{(p)}(z) = \frac{(p-1)!}{z^p} - \text{Li}_{1-p}(e^{-z}), \quad h^{(p)}(0) = \begin{cases} 1/2, & p = 1, \\ 0, & p \geq 3. \end{cases}$$

The factorial term cancels the Stirling term exactly. The remaining terms are (4.3). Taking Euler–Maclaurin and Stirling sufficiently far gives the displayed remainder; the next omitted odd power is m^{-2R-1} . All relevant derivatives are bounded on a fixed compact complex neighborhood. Uniformity follows from their remainder formulas, and differentiated versions follow by Cauchy's estimates on a slightly larger neighborhood. The formulas for E_1, E_3 follow from $B_2 = 1/6$, $B_4 = -1/30$ and $\text{Li}_0(e^{-z}) = 1/(e^z - 1)$. $\square$

The special $-z/24$ in E_1 is easy to miss. Omitting it changes the first correction and can invalidate the very small sign margin that separates the third-last and fourth-last points of a block.

5 An explicit all-orders coefficient calculus

Theorem 5.1 (Uniform bounded-offset expansion). *Fix a compact interval $I \subset (0, \infty)$ and $M > 0$. For $\alpha \in I$, $|\sigma|, |\tau| \leq M$, and integer $N = \alpha m^2 + \sigma m + \tau \geq 0$, one has, for every fixed $K \geq 0$,*

$$p_m(N) = \frac{C e^{Hm + \sigma u}}{m^2} \left(\sum_{k=0}^K \frac{D_k(\alpha; \sigma, \tau)}{m^k} + O(m^{-K-1}) \right). \quad (5.1)$$

Here u, H, C, V are as in lemma 3.1, $D_0 = 1$, and the remainder is uniform over all the indicated parameters. Each D_k is analytic in α and polynomial in σ, τ of weighted degree at most $2k$, where σ has weight 1 and τ has weight 2.

We first give a finite formula for every coefficient; the analytic proof of the expansion follows in section 6.

5.1 The Gaussian functional

Define a linear functional on polynomials in y by

$$\mathcal{G}(y^{2j}) = (-1)^j \frac{(2j)!}{2^j j!}, \quad \mathcal{G}(y^{2j+1}) = 0. \quad (5.2)$$

Equivalently, $\mathcal{G}(f) = \mathbb{E}[f(iZ)]$ for a standard normal real variable Z . The alternating sign is essential: this is a vertical saddle contour, not a real Laplace maximum with every even moment positive.

Let v be a formal variable, to be substituted by $m^{-1/2}$. All functions and derivatives in the following definition are evaluated at u unless an argument is displayed:

$$\begin{aligned} \mathcal{E}(v, y) = & \sum_{r \geq 3} \frac{F^{(r)}(u) y^r}{r! V^{r/2}} v^{r-2} + \sum_{r \geq 1} \frac{\ell^{(r)}(u) y^r}{r! V^{r/2}} v^r + \frac{\sigma y}{\sqrt{V}} v \\ & + \tau \left(u + \frac{vy}{\sqrt{V}} \right) v^2 + \sum_{r \geq 1} E_{2r-1} \left(u + \frac{vy}{\sqrt{V}} \right) v^{4r-2}. \end{aligned} \quad (5.3)$$

Then the coefficient formula is

$$D_k(\alpha; \sigma, \tau) = \mathcal{G} \left([v^{2k}] \exp \mathcal{E}(v, y) \right). \quad (5.4)$$

For fixed k , this is a finite calculation. Only the finitely many terms of $\mathcal{E}$ through degree $2k$ are needed; in particular no derivative of F beyond order $2k + 2$ occurs.

5.2 A recurrence that avoids symbolic exponentiation

Write $\mathcal{E} = \sum_{j \geq 1} e_j(y) v^j$ and $\exp \mathcal{E} = \sum_{n \geq 0} R_n(y) v^n$. Differentiating the latter identity gives

$$R_0 = 1, \quad R_n = \frac{1}{n} \sum_{j=1}^n j e_j R_{n-j}, \quad D_k = \mathcal{G}(R_{2k}). \quad (5.5)$$

This is the recurrence implemented by the included coefficient generator. Parity in (5.3) implies that R_n is odd in y when n is odd. Hence all half-integer orders vanish after applying $\mathcal{G}$. Assigning weights 1, 2 to σ, τ proves the degree bound: no parameter monomial of weight above n can occur in R_n .

5.3 The first correction in closed form

Proposition 5.2. *With every derivative evaluated at u ,*

$$\begin{aligned} D_1(\alpha; \sigma, \tau) = E_1 + \tau u - \frac{\ell'' + (\ell' + \sigma)^2}{2V} + \frac{F'''(\ell' + \sigma)}{2V^2} \\ + \frac{F''''}{8V^2} - \frac{5(F''')^2}{24V^3}. \end{aligned} \quad (5.6)$$

Proof. The first two exponent polynomials are

$$\begin{aligned} e_1(y) &= \frac{F''' y^3}{6V^{3/2}} + \frac{(\ell' + \sigma)y}{\sqrt{V}}, \\ e_2(y) &= \frac{F'''' y^4}{24V^2} + \frac{\ell'' y^2}{2V} + \tau u + E_1. \end{aligned}$$

Thus $D_1 = \mathcal{G}(e_2 + e_1^2/2)$. Substitute $\mathcal{G}(y^2) = -1$, $\mathcal{G}(y^4) = 3$, and $\mathcal{G}(y^6) = -15$. □

5.4 Quadratic subsequences

Hereafter, unless stated otherwise, $\alpha = 1$, and its argument is suppressed in D_k . For $s \in \{-1, 0, 1\}$,

$$p_m(m^2 + sm) = \frac{C e^{su} d^m}{m^2} \left(\sum_{k=0}^K D_k(s, 0) m^{-k} + O(m^{-K-1}) \right). \quad (5.7)$$

The three amplitudes are

$$\begin{aligned} C e^{-u} &= 0.070053836468553298459703821631 \dots, \\ C &= 0.158208720267250414976631099924 \dots, \\ C e^u &= 0.357296622574509427027918801595 \dots \end{aligned}$$

These agree with the existing OEIS leading constants [4, 2, 3]; the corrections are given below.

k	$D_k(-1, 0)$	$D_k(0, 0)$	$D_k(1, 0)$
1	0.278435069142632	-0.374524459843658	-1.703356625147841
2	0.163412124668439	0.073021917936842	2.310863729199545
3	0.083388796626141	-0.011224827729668	-2.892483136425176
4	0.032699494081358	0.005120280302845	3.473238845599285

Table 1: Numerical values of exact coefficients defined by (5.4). The code evaluates them at arbitrary working precision; their decimal displays are not the definitions.

6 Proof of the uniform saddle expansion

A formal local saddle expansion is not, by itself, a proof of coefficient asymptotics. The remaining contour must be bounded, uniformly in the parameters. We provide that argument explicitly.

6.1 A trigonometric estimate

Lemma 6.1. *There exist $c > 0$ and m_0 such that for $m \geq m_0$ and $|\theta| \leq \pi$,*

$$\sum_{j=\lceil m/2 \rceil}^m (1 - \cos(j\theta)) \geq cm \min(1, m^2 \theta^2). \quad (6.1)$$

Proof. When $|\theta| \leq 1/m$, use $1 - \cos x \geq c_0 x^2$ for $|x| \leq 1$ and $\sum_{j=\lceil m/2 \rceil}^m j^2 \geq c_1 m^3$. For $|\theta| \geq K/m$, a finite geometric sum gives

$$\left| \sum_{j=\lceil m/2 \rceil}^m e^{ij\theta} \right| \leq \frac{1}{|\sin(\theta/2)|} \leq \frac{\pi}{|\theta|}.$$

Choose a fixed sufficiently large K . The number of summands is at least $m/2$, so subtracting the real part of this sum gives a positive multiple of m .

For the remaining compact range $1/m \leq |\theta| \leq K/m$, put $z = m|\theta|$. Uniform Riemann-sum convergence identifies the sum divided by m with $\int_{1/2}^1 (1 - \cos zx) dx + O(1/m)$, uniformly for $1 \leq z \leq K$. This continuous integral is strictly positive for every $z > 0$: its nonnegative integrand could have zero integral only if $\cos zx = 1$ throughout an interval. Its minimum on $[1, K]$ is therefore positive. Combining the three ranges proves the claim. $\square$

6.2 Suppression away from the central saddle

Let $r_m = e^{-u/m}$. Directly from the Euler product,

$$\frac{|P_m(r_m e^{i\theta})|}{P_m(r_m)} = \prod_{j=1}^m \left(1 + \frac{2r_m^j (1 - \cos j\theta)}{(1 - r_m^j)^2} \right)^{-1/2}. \quad (6.2)$$

If α ranges over a compact subset of $(0, \infty)$, the associated u ranges over a compact subset of $(0, \infty)$. On $m/2 \leq j \leq m$ the quantities $2r_m^j/(1 - r_m^j)^2$ have positive uniform lower and finite upper bounds. Since $0 \leq 1 - \cos j\theta \leq 2$, taking logarithms in (6.2) and retaining just these factors gives

$$\frac{|P_m(r_m e^{i\theta})|}{P_m(r_m)} \leq \exp\{-cm \min(1, m^2 \theta^2)\}. \quad (6.3)$$

The coefficient integral is

$$p_m(N) = \frac{r_m^{-N}}{2\pi} \int_{-\pi}^{\pi} P_m(r_m e^{i\theta}) e^{-iN\theta} d\theta. \quad (6.4)$$

For fixed $0 < \varepsilon < 1/6$, the portion $|\theta| \geq m^{-3/2+\varepsilon}$ is bounded in modulus by $r_m^{-N} P_m(r_m) \exp(-cm^2 \varepsilon)$. The leading coefficient scale is a constant multiple of $r_m^{-N} P_m(r_m) m^{-3/2}$. Their ratio is consequently smaller than m^{-L} for every fixed L .

This estimate controls *all* the other arcs simultaneously; no unproved assumption that the positive real saddle is the only relevant contribution has been made.

6.3 Central expansion and its remainder

Write $z = u + iw$ in the equivalent vertical contour, so the integrand contains

$$\exp\{m(F(z) + \alpha z) + \sigma z + \tau z/m\} (2\pi m)^{-1/2} \exp\left\{\ell(z) + \sum_{r \geq 1} E_{2r-1}(z) m^{-(2r-1)}\right\}.$$

At $z = u$, $F'(u) + \alpha = 0$ and $F''(u) = V > 0$. Put $z = u + iZ/\sqrt{mV}$. The quadratic term is $-Z^2/2$; all remaining terms in the exponent are precisely (5.3) with $v = m^{-1/2}$ and $y = iZ$. The constant factors and the Jacobian combine to

$$\frac{e^{mH + \sigma u + \ell(u)}}{2\pi m^2 \sqrt{V}} = \frac{C e^{mH + \sigma u}}{m^2}.$$

For completeness, the usual Taylor integration can be made uniform as follows. Work first in the range $|Z| \leq c_2 m^\varepsilon$, expand through v^{2K+1} , and take proposition 4.1 to sufficiently high order. The analyticity bounds on a fixed neighborhood show that the remainder, after factoring out $e^{-Z^2/2}$, is bounded by

$$C_K m^{-K-1} (1 + |Z|^{M_K}) e^{Z^2/4}$$

for some fixed integer M_K and all sufficiently large m . Indeed the first omitted Taylor terms have polynomial growth in Z , while the nonquadratic exponential is bounded by $\exp(Cm^{-1/2}|Z|^3 + Cm^{-1/2}|Z| + \dots)$; on this range it can be absorbed into $e^{Z^2/4}$ after adjusting a constant near $Z = 0$. If necessary, reduce ε depending on K to justify each finite Taylor remainder uniformly. The bound is integrable against the remaining $e^{-Z^2/2}$.

The tails of the corresponding Gaussian polynomial integrals are superpolynomially small. Terms of odd degree in v are odd in Z and integrate to zero on the symmetric central interval. Thus termwise integration gives (5.4) and a relative $O(m^{-K-1})$ remainder. Together with (6.3), this proves theorem 5.1, including its uniformity and analyticity assertions.

7 Converting to the floor-dependent expansion

Let $t = \sqrt{N}$, $m = \lfloor t \rfloor$, and $\theta = t - m$. The exact identity

$$N = m^2 + 2\theta m + \theta^2 \tag{7.1}$$

allows us to apply theorem 5.1 with $\sigma = 2\theta$, $\tau = \theta^2$. Since $0 \leq \theta < 1$, these parameters stay in a fixed compact set. Also

$$e^{Hm + 2u\theta} = e^{Ht} e^{(2u-H)\theta} = e^{Ht} \rho^\theta. \tag{7.2}$$

Expanding $(t - \theta)^{-r-2}$ produces an exact coefficient conversion:

$$P_k(\theta) = \sum_{r=0}^k \binom{k+1}{k-r} \theta^{k-r} D_r(2\theta, \theta^2). \tag{7.3}$$

Indeed, $(1 - \theta/t)^{-r-2} = \sum_{j \geq 0} \binom{r+j+1}{j} (\theta/t)^j$, and the coefficient with $r + j = k$ has the binomial factor in (7.3). The uniform remainder is preserved because $\theta \in [0, 1]$ and $m/t \rightarrow 1$ uniformly.

This proves the expansion in theorem 2.1. The degree bound follows from the weighted degree bound for D_r : the r th summand has degree at most $k - r + 2r \leq 2k$.

7.1 The first phase polynomials

Write

$$P_1(\theta) = a + b\theta + c\theta^2, \quad (7.4)$$

where

$$\begin{aligned} a &= D_1(0, 0), \\ b &= 2 - \frac{2\ell'}{V} + \frac{F'''}{V^2}, \quad c = u - \frac{2}{V}. \end{aligned} \quad (7.5)$$

The exact rational certificate in appendix A encloses these numbers and proves the signs needed later. Numerically,

$$\begin{aligned} a &= -0.374524459843658314982205940095 \dots, \\ b &= 0.018208305709526932360906301901 \dots, \\ c &= -0.537094135888174449037648667866 \dots \end{aligned}$$

The next two polynomials are

$$\begin{aligned} P_2(\theta) &\approx 0.0730219179368424 - 0.0264113291147978\theta + 0.1866844187588720\theta^2 \\ &\quad - 0.3763383203913017\theta^3 + 0.1442350554027324\theta^4, \end{aligned} \quad (7.6)$$

$$\begin{aligned} P_3(\theta) &\approx -0.0112248277296679 + 0.0107773123530353\theta - 0.1424357689179466\theta^2 \\ &\quad + 0.1481869144022084\theta^3 - 0.2653950748358948\theta^4 + 0.1995028290093696\theta^5 \\ &\quad - 0.0258226008154378\theta^6. \end{aligned} \quad (7.7)$$

Equations (7.6)–(7.7) display rounded coefficients, not exact equalities of decimals to the mathematical coefficients; their exact definitions are (7.3) and (5.4). The data file stores longer numerical evaluations.

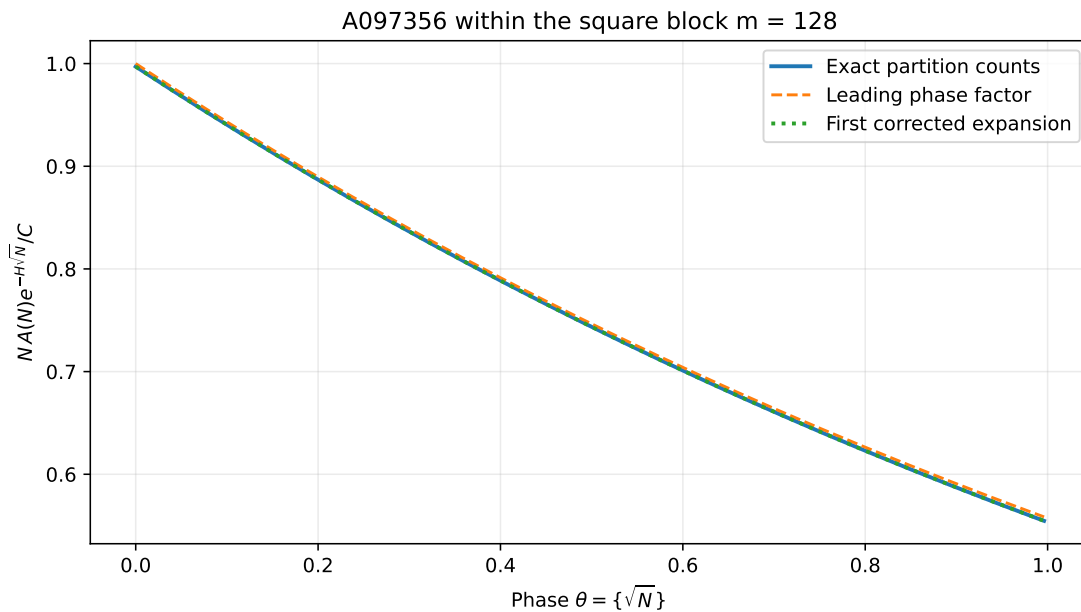


Figure 1: The normalized sequence within $128^2 \leq N < 129^2$. The exact counts and the first corrected expansion nearly coincide at this scale. The leading phase factor varies by a fixed, nonvanishing amount as the block is traversed.

7.2 Sharp envelopes and the absence of a single constant

The leading uniform formula gives

$$R(N) = C\rho^\theta(1 + O(t^{-1})). \quad (7.8)$$

Since $0 < \rho < 1$, every cluster value lies in $[C\rho, C]$. Squares have $\theta = 0$ and yield C ; the integers immediately before squares have $\theta \rightarrow 1$ and yield $C\rho$. Every intermediate phase is approached by choosing integers nearest $(m + \theta_0)^2$ for fixed $0 < \theta_0 < 1$. This proves the remaining parts of theorem 2.1.

It also proves that no constant c_0 can satisfy $A(N) \sim c_0 d^{\sqrt{N}}/N$ on the whole sequence. A power-series correction with constant coefficients cannot repair this: the missing factor ρ^θ is of order one, not a small remainder. Finally,

$$C_- = C(1 - e^{-u}) = C - Ce^{-u}. \quad (7.9)$$

Thus the conjectured lower constant is the difference of the previously recorded leading amplitudes of A206226 and A206240. This identity is a relation among amplitudes, not an assertion of a termwise identity of the integer sequences.

8 Why exactly three integers fall below the lower envelope

8.1 Strict decrease of the normalized values within each block

Lemma 8.1. *For every sufficiently large m , the function $N \mapsto R(N)$ is strictly decreasing on the integers in $[m^2, (m+1)^2 - 1]$.*

Proof. Write $N = m^2 + \sigma m$, where $0 \leq \sigma \leq 2$. The all-orders theorem with $K = 1$ and $\tau = 0$ gives, uniformly in σ ,

$$\log p_m(N) = Hm + \sigma u - 2 \log m + \log C + \frac{D_1(\sigma, 0)}{m} + O(m^{-2}).$$

When $N, N+1$ stay in the same block, σ increases by $1/m$. Since D_1 is a fixed polynomial with bounded derivative on this range,

$$\log \frac{p_m(N+1)}{p_m(N)} = \frac{u}{m} + O(m^{-2}).$$

No differentiation of an unknown remainder is used: subtracting two uniform $O(m^{-2})$ remainders remains $O(m^{-2})$. Also $\log((N+1)/N) = O(m^{-2})$ and $\sqrt{N+1} - \sqrt{N} = 1/(2m) + O(m^{-2})$. Hence

$$\log \frac{R(N+1)}{R(N)} = \frac{u - H/2}{m} + O(m^{-2}) = \frac{\log \rho}{2m} + O(m^{-2}) < 0 \quad (8.1)$$

for all sufficiently large m , uniformly within the block. □

8.2 An endpoint expansion with a decisive first correction

Proposition 8.2. *For every fixed positive integer r , let $N_{m,r} = (m+1)^2 - r$. Then*

$$\frac{R(N_{m,r})}{C_-} = 1 + \frac{\kappa_r}{m} + O(m^{-2}), \quad \kappa_r = P_1(1) + \frac{r}{2}(H - 2u). \quad (8.2)$$

In particular, $\kappa_3 < 0 < \kappa_4$.

Proof. For fixed r ,

$$t = \sqrt{(m+1)^2 - r} = m + 1 - \frac{r}{2m} + O(m^{-2}), \quad \theta = 1 - \frac{r}{2m} + O(m^{-2}).$$

Use (2.3) with $K = 1$. The phase factor relative to its lower limiting value is

$$\rho^{\theta-1} = 1 - \frac{r \log \rho}{2m} + O(m^{-2}),$$

and $P_1(\theta)/t = P_1(1)/m + O(m^{-2})$. Their product yields (8.2). The identity $-\log \rho = H - 2u$ gives its stated form. The inequalities $\kappa_3 < 0 < \kappa_4$ are certified by exact rational interval arithmetic using the analytic bounds in appendix A. The resulting enclosures are

$$\begin{aligned} -0.01618049251495 < \kappa_3 < -0.01618049246590, \\ 0.27622943999435 < \kappa_4 < 0.27622944004564. \end{aligned} \tag{8.3}$$

In particular, the negative sign of κ_3 is not being inferred from an uncertified floating-point evaluation near zero. $\square$

Proof of theorem 2.2. By proposition 8.2, for every sufficiently large m , the third-last point is strictly below C_- and the fourth-last point is strictly above C_- . By lemma 8.1, all earlier points lie above C_- , and all later points lie below C_- . This proves exactly (2.5).

For the upper envelope, the rational certificate gives $c < 0$ and $a + \max(b, 0) < 0$. Consequently $P_1(\theta) \leq -\delta$ for a fixed $\delta > 0$ throughout $0 \leq \theta \leq 1$. The uniform expansion gives

$$R(N) = C\rho^\theta(1 + P_1(\theta)/t + O(t^{-2})) < C$$

for all sufficiently large N .

There are exactly three lower-envelope violations in every sufficiently large complete block. The number of complete blocks before X is $\sqrt{X} + O(1)$, and the final incomplete block contributes at most three. Finitely many initial blocks affect only the $O(1)$ constant. Thus the violation count is $3\sqrt{X} + O(1)$. $\square$

Interpretation of the OEIS conjecture. The sharp lower *asymptotic constant* is correct. A literal eventual inequality with that exact constant is false infinitely often, but only at a set of natural density zero. Replacing C_- by any smaller positive constant gives an eventual lower inequality by the proved limit inferior. No statement here asserts that Kotešovec intended the stronger interpretation rather than an asymptotic envelope.

9 A limiting distribution for the normalized coefficients

Lemma 9.1 (Elementary phase equidistribution). *Uniformly for $0 \leq s \leq 1$,*

$$\#\{1 \leq N \leq M : \{\sqrt{N}\} \leq s\} = sM + O(\sqrt{M}). \tag{9.1}$$

Proof. In a complete block $m^2 \leq N < (m+1)^2$, the count with phase at most s is $\lfloor 2ms + s^2 \rfloor + 1$, except for the harmless endpoint adjustment at $s = 1$. It differs from $s(2m+1)$ by $O(1)$ uniformly in s . There are $O(\sqrt{M})$ complete blocks, and the final incomplete block contains $O(\sqrt{M})$ indices. Sum these estimates; removing $N = 0$ changes the result by at most one. $\square$

Theorem 9.2 (Log-uniform limiting law). *In empirical distribution over $1 \leq N \leq M$, $R(N)/C$ converges to ρ^U , where U is uniform on $[0, 1]$. Its cumulative distribution function on $[\rho, 1]$ is*

$$F_*(z) = 1 - \frac{\log z}{\log \rho}, \quad f_*(z) = -\frac{1}{z \log \rho}. \quad (9.2)$$

For every real $p \neq 0$,

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{N=1}^M R(N)^p = C^p \frac{\rho^p - 1}{p \log \rho}. \quad (9.3)$$

Proof. Equation (7.8) approximates $R(N)/C$ by $\rho^{\{\sqrt{N}\}}$ with error tending uniformly to zero outside a fixed finite initial segment. Apply lemma 9.1 to continuous test functions of the phase, then use uniform continuity to replace the phase expression by the normalized coefficients. Since ρ^U is a strictly decreasing continuous transformation of U , its distribution function is as displayed. Integrating $C^p \rho^{ps}$ over $0 \leq s \leq 1$ gives (9.3). Negative moments are allowed because the normalized sequence stays bounded away from zero for large N ; the finitely many initial positive terms have vanishing empirical weight. $\square$

The description as a distribution does not posit a random model for $A(N)$. It describes deterministic frequencies of its normalized values. The three lower-envelope exceptions per large block are compatible with this law: they have density zero and approach its lower endpoint.

10 Summation across square blocks

The unsmoothed coefficients jump at squares because the allowed part size increases. Their partial sums have a different phase law.

Theorem 10.1 (Summatory asymptotic). *Let $S(N) = \sum_{n=0}^N A(n)$, $m = \lfloor \sqrt{N} \rfloor$, and $\theta = \sqrt{N} - m$. Then uniformly in $\theta \in [0, 1)$,*

$$S(N) = \frac{C e^{Hm}}{um} \left(e^{2u\theta} - 1 + \frac{e^{2u} - 1}{e^H - 1} \right) \left(1 + O(m^{-1}) \right). \quad (10.1)$$

Proof. In block m , the leading bounded-offset formula is

$$A(m^2 + j) = \frac{C e^{Hm}}{m^2} e^{uj/m} (1 + O(m^{-1})), \quad 0 \leq j \leq 2m,$$

uniformly in j . With $J = N - m^2 = 2m\theta + \theta^2$, summing the geometric progression through $j = J$ gives

$$\sum_{j=0}^J A(m^2 + j) = \frac{C e^{Hm}}{um} (e^{2u\theta} - 1) + O(e^{Hm}/m^2). \quad (10.2)$$

This is an additive error estimate, valid even when $\theta = 0$ and its displayed leading term vanishes. A complete earlier block k contributes

$$\frac{C e^{Hk}}{uk} (e^{2u} - 1) + O(e^{Hk}/k^2).$$

For $m/2 \leq k = m - l < m$, replace m/k by $1 + O(l/m)$ and use the convergent sums $\sum_{l \geq 1} e^{-Hl}$ and $\sum_{l \geq 1} l e^{-Hl}$. This yields

$$\sum_{k < m} \sum_{n=k^2}^{(k+1)^2-1} A(n) = \frac{C e^{Hm}}{um} \frac{e^{2u} - 1}{e^H - 1} + O(e^{Hm}/m^2).$$

The blocks $k < m/2$ are exponentially smaller: the same leading bound controls all sufficiently large such k , and the remaining finite initial segment is negligible. This justifies the summation without an uncontrolled interchange of a nonuniform asymptotic series.

Add (10.2). The bracket in (10.1) is bounded below by the positive constant $(e^{2u} - 1)/(e^H - 1)$, so the additive error may be written as the stated relative error uniformly. $\square$

The leading expression glues continuously across a square. If $B = (e^{2u} - 1)/(e^H - 1)$, then $e^{2u} - 1 + B = e^H B$. Thus the limiting value at the end of one block matches the next block's leading value, after the common change in its e^{Hm} scale. This is a useful example of summation removing a jump while retaining a nonconstant phase profile.

11 Inverting the square subsequence

First remove the floor phase by considering $b(m) = p_m(m^2)$. Write $D_j = D_j(0, 0)$ and define its logarithmic coefficients by

$$\log \left(1 + \sum_{j \geq 1} D_j z^j \right) = \sum_{j \geq 1} L_j z^j. \quad (11.1)$$

A finite recurrence is

$$L_n = D_n - \frac{1}{n} \sum_{j=1}^{n-1} j L_j D_{n-j}. \quad (11.2)$$

For example $L_1 = D_1$, $L_2 = D_2 - D_1^2/2$, and $L_3 = D_3 - D_1 D_2 + D_1^3/3$.

11.1 The Lambert core and formal reversion

The dominant real equation is

$$\frac{C e^{HX}}{X^2} = Y, \quad HX - 2 \log X = \log Y - \log C. \quad (11.3)$$

For $X > 2/H$ it is strictly increasing and has the inverse (2.6), using W_{-1} , not W_0 . The principal branch would select the small solution of the same algebraic rearrangement and is not the desired asymptotic inverse.

Proposition 11.1 (All-orders inverse coefficients). *The formal inverse of the square-subsequence expansion has the unique form*

$$\widehat{x}(Y) = X(Y) + \sum_{r \geq 1} q_r X(Y)^{-r}. \quad (11.4)$$

If $\Delta_{r-1}(X) = \sum_{j=1}^{r-1} q_j X^{-j}$, then

$$q_r = -\frac{1}{H} [X^{-r}] \left\{ -2 \log \left(1 + \frac{\Delta_{r-1}}{X} \right) + \sum_{j=1}^r L_j (X + \Delta_{r-1})^{-j} \right\}. \quad (11.5)$$

In particular,

$$\begin{aligned} q_1 &= -L_1/H, \\ q_2 &= -2L_1/H^2 - L_2/H, \\ q_3 &= (2q_2 + L_1 q_1 - L_3)/H. \end{aligned} \quad (11.6)$$

Proof. Put $x = X + \Delta$ in $Hx - 2 \log x + \sum_{j \geq 1} L_j x^{-j} = \log Y - \log C$ and subtract (11.3). The resulting equation is

$$H\Delta - 2 \log(1 + \Delta/X) + \sum_{j \geq 1} L_j (X + \Delta)^{-j} = 0.$$

At order X^{-r} the previously unknown coefficient occurs only as Hq_r , since all its occurrences in the logarithm and inverse powers carry at least one additional negative power of X . This proves both uniqueness and the recurrence. Equating the first three coefficients gives (11.6). $\square$

Numerically the first coefficients are

$$\begin{aligned} q_1 &= 0.169152574435079450236784971776 \dots, \\ q_2 &= 0.151490069409063238951250698423 \dots, \\ q_3 &= 0.108853991540605915550641858907 \dots, \\ q_4 &= 0.058606290184968379522730931714 \dots \end{aligned}$$

To express the same expansion using only iterated logarithms, put $\Lambda = \log(Y/(CH^2))$. Reversion of $HX - 2 \log(HX) = \Lambda$ and inclusion of the first correction give

$$\widehat{x}(Y) = \frac{1}{H} \left(\Lambda + 2 \log \Lambda + \frac{4 \log \Lambda - HL_1}{\Lambda} + O\left(\frac{(\log \Lambda)^2}{\Lambda^2}\right) \right). \quad (11.7)$$

The Lambert form keeps the entire leading power-logarithmic core unexpanded and is usually more accurate at a given truncation order.

11.2 What a real inverse means for an integer sequence

The integer data $b(m)$ do not specify a canonical smooth interpolation. To state a rigorous analytic meaning without inventing one, fix K and use the explicit smooth approximation

$$B_K(x) = \frac{Ce^{Hx}}{x^2} \left(1 + \sum_{j=1}^K D_j x^{-j} \right). \quad (11.8)$$

It is positive and strictly increasing for sufficiently large x , since $(\log B_K)'(x) = H - 2/x + O(x^{-2})$. Let $x_K(Y)$ denote its unique large inverse. The formal calculation and the mean value theorem show

$$x_K(Y) = X(Y) + \sum_{j=1}^K q_j X(Y)^{-j} + O(X(Y)^{-K-1}). \quad (11.9)$$

Indeed substitution of the right-hand truncation leaves a logarithmic residual of order X^{-K-1} , and the derivative of the logarithmic left-hand side is bounded away from zero near the root.

Proposition 11.2 (Transport to a discrete threshold). *Let $M(Y) = \min\{m \geq 0 : b(m) \geq Y\}$. There is a constant $c_K > 0$ such that, for all sufficiently large Y , with $\delta_K = c_K x_K(Y)^{-K-1}$,*

$$\lceil x_K(Y) - \delta_K \rceil \leq M(Y) \leq \lceil x_K(Y) + \delta_K \rceil. \quad (11.10)$$

The same type of brackets holds with x_K replaced by the explicit truncation in (11.9), after enlarging c_K .

Proof. At integer m , theorem 5.1 gives $\log b(m) - \log B_K(m) = O(m^{-K-1})$. Near x_K , the logarithmic derivative of B_K is at least $H/2$ for large Y . Displacing the argument by a sufficiently large constant times x_K^{-K-1} therefore makes the sign of $\log b(m) - \log Y$ certain on either side of the bracket. Monotonicity of $b(m)$ excludes more distant threshold values. Taking ceilings yields (11.10). The real approximation error (11.9) has the same order and can be absorbed by increasing the constant. $\square$

A ceiling cannot in general be pulled through an $O(\cdot)$ term. An exact formula $M(Y) = \lceil x_K(Y) \rceil$ follows from (11.10) only under a separation condition that ensures both endpoint ceilings agree, for example $\text{dist}(x_K(Y), \mathbb{Z}) > \delta_K$. Arbitrarily small real errors can change a ceiling near an integer. This is the reason to keep the real inverse and the staircase statement distinct, as in ProveIt's remainder-transport and staircase organization [10].

[Write note, batch 7301, 1 October 2026.] Proposition 11.2 and the separation remark after it are not new. They are the special case $x_* = x_K(Y)$, $\eta = \delta_K$ of the staircase recovery theorem of ProveIt's transseries volume [11], Theorem `p0:thm:staircase`. Once the real inverse ν of a strictly increasing interpolation of b is known to lie within δ_K of $x_K(Y)$, its part (1), exact rounding $M(Y) = \lceil \nu(Y) \rceil$, and the monotonicity of the ceiling give the bracket (11.10), and its part (2), the separation condition $\lceil x_* - \eta \rceil = \lceil x_* + \eta \rceil$, gives the exact formula $M(Y) = \lceil x_K(Y) \rceil$. The order-theoretic content of part (2) is formalized in Lean as `Fabius.staircase_separation` (with `Fabius.staircase_cceil` for part (1)) in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean` [12]. What is specific to A097356 here is the analytic input: the all-orders estimate $\log b(m) - \log B_K(m) = O(m^{-K-1})$ of theorem 5.1 that produces the width δ_K . That input is not formalized, and this report gains no formal status from the Lean file. The proposition is printed with its delivered proof as a specialization, not as a new result.

For $p_m(m^2 + sm)$ with $s = \pm 1$, replace C by Ce^{su} and D_j by $D_j(s, 0)$. All the same reversion and threshold arguments apply.

12 Inverting the complete floor-dependent sequence

12.1 Monotonicity and the geometry of a square block

The sequence $A(N)$ is nondecreasing: append a part of size one to a partition counted at N ; the resulting partition is allowed at $N + 1$ because the cutoff cannot decrease. The map is injective. Since $A(m^2) \sim Cd^m/m^2 \rightarrow \infty$, the inverse threshold $Q(Y)$ is finite.

For a real phase $s \in [0, 1)$ and integer points with $\sqrt{N} = m + s$, the leading logarithm is

$$\log A(N) = \log C + Hm + 2us - 2 \log m + O(m^{-1}). \quad (12.1)$$

As a block is traversed, this logarithm increases by about $2u$; at the next square, the leading jump is $H - 2u = -\log \rho > 0$. Thus a positive fraction of large target logarithms lie in gaps crossed at a single square. A smooth phase-blind inverse cannot see these plateaus.

12.2 Proof of the global inverse approximation

Proof of theorem 2.3. The core $X = X(Y)$ satisfies $\log Y = \log C + HX - 2 \log X$. Uniform bounds on the phase factor in (2.3) first imply

$$\sqrt{Q(Y)} = X + O(1). \quad (12.2)$$

For example, changing t by a sufficiently large fixed constant changes $Ht - 2 \log t$ by more than the bounded phase correction, and comparison with the nearest available integer index brackets the threshold.

It is therefore enough to consider $t = X + O(1)$, where $\log \lfloor t \rfloor = \log X + O(X^{-1})$. Define the right-continuous increasing comparison function

$$g_X(t) = \log C - 2 \log X + H \lfloor t \rfloor + 2u \{t\}. \quad (12.3)$$

At the available grid points $t = \sqrt{N}$ near X , (12.1) gives $\log A(N) = g_X(t) + O(X^{-1})$, uniformly. The generalized real inverse of g_X at $\log Y$ is exactly $T(X)$: if $X = k + \eta$, then the target increment in block k is $H\eta$; it is attained at phase η/γ when $\eta \leq \gamma$, and otherwise the first crossing is at $k + 1$.

The map T is continuous and nondecreasing, with slopes $1/\gamma$ and 0 on its linear pieces. The generalized inverse of g_X is therefore globally Lipschitz as a function of the target logarithm, with constant $1/(2u)$. A perturbation of that logarithm by $O(X^{-1})$ displaces its inverse by $O(X^{-1})$, even at the beginning or end of a plateau. The grid spacing $\sqrt{N+1} - \sqrt{N}$ near X is also $O(X^{-1})$. Comparing the target with $g_X \pm c/X$ and then moving to the neighboring grid points proves

$$\sqrt{Q(Y)} = T(X) + O(X^{-1}).$$

Since $T(X) = X + O(1)$, squaring changes the error to $O(1)$, as stated. $\square$

Write

$$\psi(\eta) = \min(\eta/\gamma, 1) - \eta = \begin{cases} (1/\gamma - 1)\eta, & 0 \leq \eta \leq \gamma, \\ 1 - \eta, & \gamma \leq \eta \leq 1. \end{cases} \quad (12.4)$$

Then $T(X) = X + \psi(\{X\})$, and the global inverse theorem gives

$$Q(Y) - X(Y)^2 = 2X(Y)\psi(\{X(Y)\}) + O(1). \quad (12.5)$$

For a fixed phase $\eta_0 \in (0, 1)$, set $Y_k = Ce^{H(k+\eta_0)}/(k+\eta_0)^2$. Then $X(Y_k) = k + \eta_0$ and $\psi(\eta_0) > 0$. Along these targets the omitted term has size $\Theta(X)$. Rounding Y_k to a nearby integer changes X by an exponentially small amount and does not alter this conclusion. The phase-blind error is thus genuinely of order $\log Y$, not just a loose upper bound or a one-integer rounding issue.

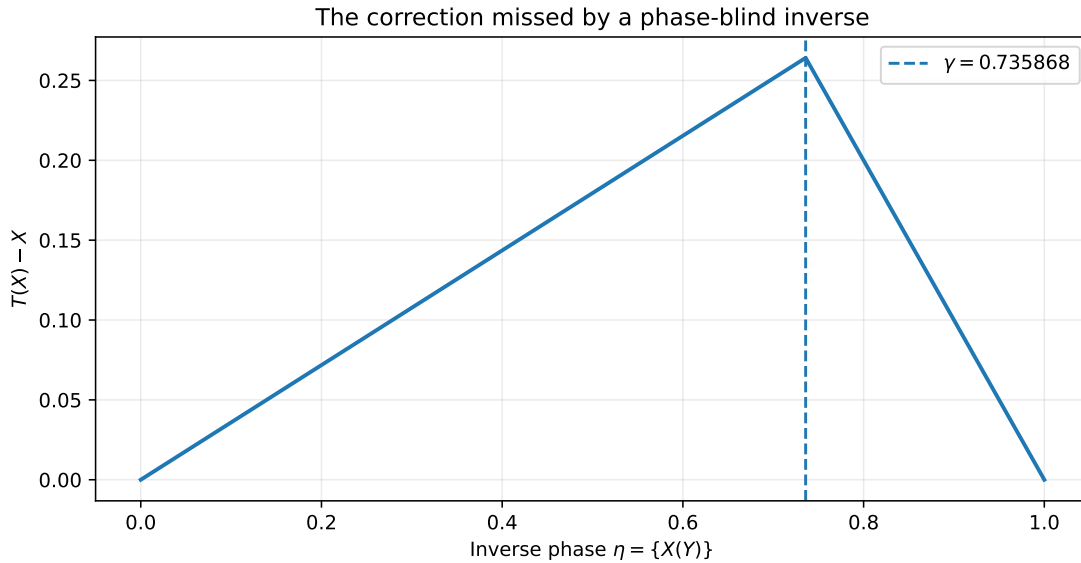


Figure 2: The tent-shaped correction $\psi(\eta) = T(X) - X$. Its maximum is $1 - \gamma = 0.264131709289 \dots$. The flat pieces of T , not of ψ , correspond to target ranges whose inverse is the next perfect square.

12.3 Exact plateaus away from the transition layers

Proposition 12.1. *Fix a compact interval $J \subset (\gamma, 1)$. If $X(Y) = k + \eta$ with $\eta \in J$, then for all sufficiently large k ,*

$$Q(Y) = (k + 1)^2. \quad (12.6)$$

Proof. At the last index of block k ,

$$\log A((k + 1)^2 - 1) = \log C + Hk + 2u - 2 \log k + O(k^{-1}).$$

At the next square,

$$\log A((k+1)^2) = \log C + H(k+1) - 2\log(k+1) + O(k^{-1}).$$

The target logarithm is $\log C + Hk + H\eta - 2\log k + O(k^{-1})$. On J , both $H\eta - 2u$ and $H(1 - \eta)$ have positive lower bounds. Thus the target lies strictly between these two adjacent values for large k . Monotonicity proves the result. $\square$

12.4 The first interior correction

Fix instead a compact interval $J \subset (0, \gamma)$. Let $X = k + \eta$ with $\eta \in J$, and put

$$\sigma_0(\eta) = \frac{H\eta}{u}, \quad \tau_0(\eta) = \frac{-2\eta - D_1(\sigma_0(\eta), 0)}{u}. \quad (12.7)$$

Consider the real approximation

$$n_*(Y) = k^2 + \sigma_0(\eta)k + \tau_0(\eta). \quad (12.8)$$

It is useful for locating the integer threshold even though no canonical interpolation of $A(N)$ is being specified.

Proposition 12.2 (Interior threshold brackets). *Uniformly for $\eta \in J$, there is a constant $c_J > 0$ such that for all sufficiently large k ,*

$$\lceil n_*(Y) - c_J/k \rceil \leq Q(Y) \leq \lceil n_*(Y) + c_J/k \rceil. \quad (12.9)$$

In particular $Q(Y) = \lceil n_(Y) \rceil$ whenever the endpoint ceilings in (12.9) agree.*

Proof. For $N = k^2 + \sigma_0 k + \tau$, theorem 5.1 gives

$$\log p_k(N) = \log C + Hk - 2\log k + u\sigma_0 + \frac{u\tau + D_1(\sigma_0, 0)}{k} + O(k^{-2}).$$

The target has expansion

$$\log Y = \log C + Hk - 2\log k + H\eta - \frac{2\eta}{k} + O(k^{-2}).$$

Matching its constant and k^{-1} terms gives (12.7). The slope of the logarithmic asymptotic approximation with respect to N is $u/k + O(k^{-2}) > 0$. Hence a logarithmic residual $O(k^{-2})$ gives a real index displacement $O(k^{-1})$. Equivalently, evaluating at integer points a sufficiently large constant times $1/k$ on either side of n_* makes the signs certain. The compact separation from 0 and γ keeps these points in block k and away from its jumps. Monotonicity and ceilings yield (12.9). $\square$

12.5 An all-orders interior recurrence

Let

$$\log \left(1 + \sum_{j \geq 1} D_j(\sigma, 0) w^j \right) = \sum_{j \geq 1} L_j(\sigma) w^j. \quad (12.10)$$

Every $L_j(\sigma)$ is a polynomial obtained from (11.2). Define a formal series $\sigma(w, \eta) = \sum_{r \geq 0} \sigma_r(\eta) w^r$, beginning with $\sigma_0 = H\eta/u$. The equation to be solved is

$$u\sigma(w, \eta) + \sum_{j \geq 1} L_j(\sigma(w, \eta)) w^j = H\eta - 2\log(1 + \eta w). \quad (12.11)$$

If $\sigma^{<r}(w, \eta) = \sum_{v=0}^{r-1} \sigma_v(\eta) w^v$, the next coefficient is explicitly

$$\sigma_r(\eta) = \frac{1}{u} \left\{ \frac{2(-1)^r \eta^r}{r} - [w^r] \sum_{j=1}^r L_j(\sigma^{<r}(w, \eta)) w^j \right\}, \quad r \geq 1. \quad (12.12)$$

The unknown σ_r occurs only through $u\sigma_r$ at order w^r ; this proves formal uniqueness. At $r = 1$ it gives precisely τ_0 . For a truncation through $r = R$ put

$$n_R(Y) = k^2 + k \sum_{r=0}^R \sigma_r(\eta) k^{-r}.$$

The analytic remainder argument used in proposition 12.2, now with theorem 5.1 at higher order, yields uniform brackets

$$\left[n_R(Y) - c_{J,R} k^{-R} \right] \leq Q(Y) \leq \left[n_R(Y) + c_{J,R} k^{-R} \right]. \quad (12.13)$$

Indeed the logarithmic residual is $O(k^{-R-1})$ and the local logarithmic slope is comparable to $1/k$. This proves an all-orders inverse expansion on every compact interior phase interval.

No all-orders assertion is made here uniformly through $\eta = 0$ or $\eta = \gamma$. Those switching layers require their own scaled coordinates. The coarse global formula (2.8) *does* remain uniform through both transitions.

13 Exact and high-precision computational checks

13.1 What was computed

The included run used Python with mpmath 1.3.0 at 80 decimal digits. The counts themselves were not computed by floating point. They were generated by the exact recurrence that adds the allowed parts one at a time:

$$p_m(N) = p_{m-1}(N) + p_m(N - m), \quad (13.1)$$

with $p_0(0) = 1$, $p_0(N) = 0$ for $N > 0$, and zero values at negative arguments. In-place updates run over increasing N for each new m . The final run produced all A097356 values through $N = 66048$ and the three quadratic subsequences through $m = 256$.

As transcription checks, the first 54 terms of A097356 and first 17 terms of A206226 were compared with the inspected OEIS entries [1, 2]. All agreed. All 66048 consecutive monotonicity comparisons passed. The coefficient generator was independently checked against the closed formula (5.6) at nine choices of σ . The exact rational checks described in appendix A also passed. These are finite checks of the supplied implementation and specific signs; they do not replace the all-orders analytic proof.

13.2 Square-subsequence accuracy

Table 2 records relative errors ($\text{approximation}/p_m(m^2) - 1$) using the truncation through D_K . The rapid improvement is consistent with the successive proved powers of $1/m$.

A single small- m accuracy pattern does not prove convergence of the asymptotic series or identify secondary saddles. For example, the $m = 16$ row need not follow the eventual algebraic error ratios as cleanly as the larger rows. No inference about exponentially small sectors is based on that observation.

13.3 The third-last point is a subtle case

The table below shows $R((m+1)^2 - r)/C_- - 1$ near the end of a block.

m	$K = 0$	$K = 1$	$K = 2$	$K = 3$	$K = 4$
16	$2.37 \cdot 10^{-2}$	$-2.89 \cdot 10^{-4}$	$2.71 \cdot 10^{-6}$	$-9.65 \cdot 10^{-8}$	$-1.65 \cdot 10^{-8}$
32	$1.18 \cdot 10^{-2}$	$-7.18 \cdot 10^{-5}$	$3.42 \cdot 10^{-7}$	$-4.88 \cdot 10^{-9}$	$5.60 \cdot 10^{-11}$
64	$5.87 \cdot 10^{-3}$	$-1.79 \cdot 10^{-5}$	$4.28 \cdot 10^{-8}$	$-3.05 \cdot 10^{-10}$	$1.74 \cdot 10^{-12}$
128	$2.93 \cdot 10^{-3}$	$-4.46 \cdot 10^{-6}$	$5.35 \cdot 10^{-9}$	$-1.91 \cdot 10^{-11}$	$5.42 \cdot 10^{-14}$
256	$1.46 \cdot 10^{-3}$	$-1.12 \cdot 10^{-6}$	$6.69 \cdot 10^{-10}$	$-1.19 \cdot 10^{-12}$	$1.69 \cdot 10^{-15}$

Table 2: Relative errors for A206226. The CSV files retain more digits. Here $a \cdot 10^{-j}$ means $a \times 10^{-j}$.

m	$r = 1$	$r = 2$	$r = 3$	$r = 4$
16	-0.03435127	-0.01580701	0.003070873	0.02228763
32	-0.01793700	-0.008717003	0.000586331	0.009973718
64	-0.009173963	-0.004582366	0.0000300765	0.004663456
128	-0.004640419	-0.002350006	-0.0000543790	0.002246474
256	-0.002333841	-0.001190087	-0.0000450293	0.001101334

Table 3: The third-last index changes sign later than the last two. The eventual theorem rests on the certified $\kappa_3 < 0$, not on extrapolation from these samples.

The four limiting coefficients in (8.2) are approximately

$$(\kappa_1, \kappa_2, \kappa_3, \kappa_4) = (-0.60100036, -0.30859043, -0.01618049, 0.27622944).$$

In particular, the small absolute size of κ_3 explains why an initial data inspection could miss the eventual third exception.

13.4 Inverse accuracy and the plateaus

For the inverse tests, target values were chosen as integers near $Ce^{H(k+\eta)}/(k+\eta)^2$; thresholds were then found by binary search in the exact integer table. Table 4 gives the block $k = 128$ results. “Interior ceiling” means the value $\lceil n_*(Y) \rceil$ from (12.8); it is only used on interior phases.

η	Exact $Q(Y)$	$X^2 - Q(Y)$	$T(X)^2 - Q(Y)$	Interior ceiling error
0.10	16420	-10.3900	-1.1927	0
0.30	16490	-29.1100	-1.4673	0
0.50	16560	-47.7500	-1.5942	0
0.70	16631	-67.3100	-2.5733	0
0.80	16641	-51.5600	0	plateau
0.95	16641	-12.8975	0	plateau

Table 4: An ordinary smooth inverse misses dozens of indices; the phase map has a bounded error, and the first interior correction locates the exact integer thresholds in these tests. Exact rounding in arbitrary cases still requires the separation condition in the theorem.

13.5 Reproduction and interpretation of the outputs

From the directory containing the scripts, run

```
python certificate.py
python verify.py --max-m 256 --output results
```

The first command uses only the Python standard library and performs exact rational interval computations. The second needs `mpmath` and reruns the coefficient evaluation and integer checks. Its time cost is dominated by $O(m_{\max}^3)$ arbitrary-precision integer additions; storage is $O(m_{\max}^2)$ integers. These are operation counts, not constant-time bit-complexity bounds.

The data directory contains coefficient JSON files and separate CSV files for quadratic expansions, floor expansions, endpoint values, the last-four classification, inverse thresholds, and summatory approximations. `verification_summary.json` records the completed run. The figures use a saved subset of the exact-count comparison data; they are illustrations, not substitutes for the numerical tables or proofs.

[Write note, batch 7301, 1 October 2026.] In the collection the four scripts are in `code/`, and the eleven files of the delivered `results/` directory, with the two requirements files, are in `data/`. Run the commands above from a copy of `code/` with an explicit output directory, for example `-output` pointing to a scratch directory, never to the shipped `data/`. The script writes its CSV files with CRLF line ends on every platform, as delivered, and on Windows also writes its JSON files with CRLF ends, so a rerun there differs from the shipped JSON files in line ends only. The figure script reads `results/plot_data.json` and writes `figures/` relative to the working directory by default; pass `-data` and `-output` explicitly. At intake both commands were rerun on a copy: of the ten files the driver writes, the six CSV files and `plot_data.json` were byte-identical to the delivery and the other three JSON files equal apart from line ends; its standard output equals the shipped `data/run_log.txt`, which is a capture of that output and not a file the script writes.

14 Further research questions and a formalization route

The results suggest several nontrivial continuations. The following are proposals, not claims that their conclusions have been proved here.

14.1 Effective determination of the first stable three-exception block

The analytic proof gives a finite but unspecified m_0 . An effective version would bound the m^{-2} remainder in the endpoint formula and in the block-ratio estimate, then combine it with certified computation for the finite initial range. The small margin $|\kappa_3| \approx 0.01618$ makes a crude uniform bound potentially inefficient. A useful target is the *least* m_0 such that exactly three exceptions occur in every block $m \geq m_0$, together with a proof that no later reversals occur. Uncertified testing of a large interval would not settle the latter requirement.

14.2 All-orders switching layers for the inverse

The global inverse error is bounded, and the interior inverse has all orders, but the transition regions have only been covered at coarse order. Resolve the layers $\eta = O(1/k)$ and $\eta - \gamma = O(1/k)$ by comparing corrected block endpoints and square values. A complete formulation should give the shifted jump thresholds, integer ceiling brackets, and a uniform matching rule between interior roots and exact plateaus. This would extend the phase-sensitive inverse from an atlas with excluded transition layers to a uniformly corrected global description.

14.3 Exponential sectors and optimal truncation

The minor-arc proof only shows contributions smaller than every fixed algebraic order. It does not identify their leading exponential scales. A genuinely exponentially improved expansion should identify the relevant rational-angle or complex saddles, compare their actions, and describe any changes of

dominance. One should also study the growth of D_k as $k \rightarrow \infty$, whether a Gevrey bound holds, and how optimal truncation relates to the first omitted saddle. Calling the present Poincaré series a complete resurgent transseries would skip these unresolved steps.

14.4 Other moving cutoffs and their endpoint defect counts

For $A_c(N) = p_{\lfloor c\sqrt{N} \rfloor}(N)$ with fixed $c > 0$, the saddle has $\alpha = c^{-2}$. The current uniform theorem supplies the analytic input for a related phase calculus, but the arithmetic block endpoints need not be integer squares. Determine whether the number of points violating a sharp asymptotic lower envelope stabilizes, becomes periodic, or has a phase-dependent distribution. The sign of a first endpoint correction can change with c , producing transitions in that count. Rational and irrational c^2 may lead to genuinely different endpoint arithmetic.

14.5 Two simultaneous restrictions

Partitions constrained both in largest part and number of parts are encoded by Gaussian binomial coefficients. Jiang and Wang provide a generalized Hardy–Ramanujan formula in a two-restriction regime [8]. A natural continuation is a two-phase all-orders calculus when both boundaries involve floors of multiples of $\sqrt{N}$. The inverse could then have interacting plateau mechanisms rather than a single tent correction. The one-dimensional proof should not be assumed to establish this extension without new uniform saddle and boundary estimates.

14.6 Uniform limits as the saddle moves to zero or infinity

Our theorem is uniform only for α in compact subsets of $(0, \infty)$. The limits $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$ move the saddle to a boundary of this range. A useful next problem is to match the present expansions with other partition regimes and determine how many derivatives and remainder constants remain controlled when $\alpha = \alpha(m)$ varies. Existing leading formulas do not by themselves justify termwise passage of all our corrections to those boundary regimes.

14.7 All-orders summation and arithmetic statistics

The summatory formula was proved only to relative order $1/m$. Applying Euler–Maclaurin to each finite block and then summing the exponentially weighted previous blocks should produce higher phase corrections. Explicitly derive them, with uniform control at $\theta = 0$, and study whether successive integrations remove higher-order phase irregularities. Another direction is an effective error term for the limiting distribution of $R(N)$, including the deterministic displacement of its endpoints and the sparse lower-envelope violations.

14.8 A modular Lean formalization

A practical route separates algebraic, finite, and analytic components. The repository’s documented transseries modules already distinguish coefficient identities, asymptotic scales, inverse cores, and discrete thresholds [10]. A corresponding development here would proceed through the following dependency layers:

1. Formalize the exact finite coefficient operator and (5.5), the weighted-degree bound, and the conversion (7.3). These are finite polynomial identities, independent of contour analysis.
2. Formalize the rational sign certificate, including the Bernoulli and logarithmic remainder bounds. This certifies the endpoint signs but not yet the partition asymptotic.

3. Establish the analytic Euler–Maclaurin product expansion, the uniform trigonometric bound, and the saddle integration theorem with explicit parameter domains.
4. Derive the floor expansion, the block monotonicity, and the three-exception theorem from those analytic estimates and finite signs. Then formalize the generalized inverse comparison and ceiling brackets.

This division makes the missing obligations explicit. The supplied Python program is not a substitute for a kernel-checked proof, and no current theorem is labeled machine-formalized in this report.

[Write note, batch 7301, 1 October 2026.] The last order-theoretic step of item 4, passing from a real approximation of the inverse to the integer threshold under the separation condition, already exists in the repository: `Fabius.staircase_separation` and `Fabius.staircase_ceil` in `StaircaseInv` `ersion.lean` [12] formalize parts (1)–(2) of the transseries volume’s staircase theorem (see the note after proposition 11.2). Every analytic obligation of items 1–4 remains open, and no statement of this report is formalized.

15 Conclusion

The natural partition sequence A097356 has a rigid asymptotic structure: one positive real saddle, an order-one floor phase, polynomial corrections, and a staircase inverse with macroscopic phase effects. The conjectured lower constant is exactly right as a limit inferior, but the associated literal lower inequality has exactly three violations before every sufficiently large square. The distinction only becomes visible after retaining the first correction and proving its endpoint signs.

The analysis yields more than an isolated constant evaluation. One finite coefficient operator handles a compact family of quadratic restrictions, recovers the existing OEIS amplitudes, generates all algebraic correction orders, and supports both smooth and discrete inverse statements. It also exposes the limits of the result: effective cutoffs, switching layers, exponentially small sectors, and formal verification remain separate research tasks rather than hidden assumptions.

A Exact rational sign certificates

The delicate claim $\kappa_3 < 0$ deserves a reproducible certificate whose validity does not depend on trusting a floating-point root finder. This appendix supplies the analytic bounds used by `certificate.py`; the script then performs only exact rational arithmetic.

A.1 Power series with controlled derivatives

For $|u| < 2\pi$, the Bernoulli expansion gives

$$F(u) = 1 - \log u + \frac{u}{4} - \sum_{r \geq 1} \frac{B_{2r} u^{2r}}{(2r)(2r)!(2r+1)}, \quad (\text{A.1})$$

$$\ell(u) = \frac{u}{4} - \sum_{r \geq 1} \frac{B_{2r} u^{2r}}{2(2r)(2r)!}, \quad (\text{A.2})$$

$$E_1(u) = -\frac{1}{12} - \frac{1}{12} \sum_{r \geq 1} \frac{B_{2r} u^{2r}}{(2r)!}. \quad (\text{A.3})$$

These follow from the generating function $u/(e^u - 1) = \sum_{n \geq 0} B_n u^n / n!$ by integration and the identity $E_1 = -u \coth(u/2)/24$.

The classical Bernoulli–zeta identity, equivalently the Bernoulli Fourier series, is [9]

$$\frac{|B_{2r}|}{(2r)!} = \frac{2\zeta(2r)}{(2\pi)^{2r}} < \frac{4}{6^{2r}}. \quad (\text{A.4})$$

Here $\zeta(2r) < 2$ and $\pi > 3$ suffice; high-precision values of either constant are unnecessary. On $4/5 \leq u \leq 1$, the absolute value of any of the first four derivatives of a tail monomial in (A.1)–(A.3) is at most $4(2r)^4/6^{2r}$, after discarding the additional denominators that would only improve the estimate. For a truncation through $r = R = 16$, the ratio of successive such majorants is less than $1/2$. Therefore the omitted series and each derivative through order four are bounded in absolute value by the explicit rational number

$$\varepsilon_R = \frac{8(2R+2)^4}{6^{2R+2}}, \quad R = 16. \quad (\text{A.5})$$

The elementary terms involving $\log u$ and powers of u^{-1} are handled separately and exactly as interval expressions.

For $\log u$, put $w = (1 - u)/(1 + u)$. Since $u \in [4/5, 1]$, $0 \leq w \leq 1/9$. For $T = 24$,

$$\log u = -2 \sum_{j=0}^{T-1} \frac{w^{2j+1}}{2j+1} + \epsilon_T, \quad |\epsilon_T| \leq \frac{2w^{2T+1}}{(2T+1)(1-w^2)}. \quad (\text{A.6})$$

This follows by bounding the positive tail of the series for $\operatorname{arctanh} w$. All evaluations use rational interval endpoints, with the error bounds widened outward.

A.2 Bracketing the saddle

Set

$$u_- = \frac{814651136747}{10^{12}}, \quad u_+ = \frac{814651136748}{10^{12}}. \quad (\text{A.7})$$

Evaluate $J = -F'$ from (A.1), with its derivative remainder bounded by (A.5). Exact rational arithmetic gives

$$\begin{aligned} 0.00000000000090 &< J(u_-) - 1 < 0.00000000000091, \\ -0.00000000000058 &< J(u_+) - 1 < -0.00000000000057. \end{aligned} \quad (\text{A.8})$$

Since J is strictly decreasing by lemma 3.1, its root lies in $[u_-, u_+]$. The use of a rational root bracket, rather than a decimal root asserted to be accurate, is what permits an exact finite certificate.

A.3 Propagating the interval through the formulas

The script evaluates F , its first four derivatives, ℓ' , ℓ'' , and E_1 on $[u_-, u_+]$, adding the rigorous series-tail bounds. It then substitutes these intervals into (5.6) and (7.5). Division is only performed when the denominator interval is separated from zero; in particular the interval for V is positive. The results below have been rounded outward for display.

The code also checks the simple inequalities

$$a + \max(b, 0) < 0, \quad c < 0, \quad \kappa_3 < 0 < \kappa_4.$$

They are the complete finite sign input to the upper-envelope and three-exception theorems. No unexplained approximate evaluation of π , a polylogarithm, or an exponential occurs in this certificate: $H = u + F(u)$ at $\alpha = 1$ and all other quantities are evaluated from the bounded power series above.

Quantity	Certified lower endpoint	Certified upper endpoint
H	2.21412213851459	2.21412213851710
V	1.47956870313163	1.47956870313535
a	-0.37452445985485	-0.37452445983240
b	0.01820830570211	0.01820830571759
c	-0.53709413589011	-0.53709413588571
κ_1	-0.60100035753254	-0.60100035749000
κ_3	-0.01618049251495	-0.01618049246590
κ_4	0.27622943999435	0.27622944004564

Table 5: Outward-rounded rational interval enclosures. Each displayed bound is weaker than the exact rational inequality checked by the code.

Trust boundary. This is an exact-arithmetic certificate with a written analytic justification. Its implementation uses Python’s arbitrary-precision integers and `fractions.Fraction`. It is independently rerunnable, but it has not been proved correct inside a proof assistant. The saddle and contour arguments remain conventional mathematical proofs.

B Artifact inventory and source audit

B.1 Package contents

File or directory	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Editable source and compiled report.
<code>partition_asymptotics.py</code>	Saddle constants, all-orders coefficient operator, phase polynomials, inverse coefficients, exact partition tables.
<code>certificate.py</code>	Standard-library-only rational interval proof checks.
<code>verify.py</code>	Reproduction driver for exact counts and numerical comparisons.
<code>make_figures.py</code>	Recreates the two figures from the saved data.
<code>results/</code>	JSON coefficients, sign enclosures, CSV comparisons, run summary.
<code>figures/</code>	Two PDF figures and PNG previews.
<code>README.md</code>	Build and reproduction instructions; limitations.
<code>PROVENANCE.md</code>	Repository checkpoint, consulted sources, attribution scope.
<code>requirements.txt</code>	Numerical dependency used in the completed run.

To rebuild the document, run `latexmk -pdf article.tex` in this directory, or run `pdflatex article.tex` repeatedly until the references stabilize. The PDF figures are included so that rebuilding the article does not require recalculating the integer tables.

[Write note, batch 7301, 1 October 2026.] The table above describes the delivered package. In the collection: `article.pdf` is a build of this edition, not the delivered PDF; the three Python modules and `make_figures.py` are in `code/`; the contents of `results/` and both requirements files (`requirements.txt` and `requirements-figures.txt`, the latter not listed above) are in `data/`; `figures/`, `README.md`, `PROVENANCE.md` and `VALIDATION.md` (also not listed above) are at the report root, and `README.md` is the collection’s report guide, not the delivered README. The delivered checksum ledger is not shipped. Appendix C gives the full map.

B.2 Source and novelty audit

The public OEIS entries were checked on October 1, 2026. The source audit records the actual scope of the reading: A097356 and its conjectured constant, the three quadratic restricted-partition entries, A258268, Canfield’s and Romik’s treatments of Szekeres’ asymptotic, and the repository’s transseries

documentation. The wider two-restriction work of Jiang–Wang is cited as a research continuation, not as a theorem used silently in the proof. The Bernoulli Fourier identity is used explicitly for the certificate tail bound.

A repository search and the inspected recursive tree did not locate an A097356-specific item at the checked checkpoint. This does not establish that no related argument exists somewhere in its broader corpus or in unindexed literature. The report is consequently framed as a proved solution of a precise posted asymptotic question and a set of explicit refinements, rather than as a certified claim of first discovery of restricted-partition asymptotics.

C Provenance of the collection edition

[Write note, batch 7301, 1 October 2026.] This appendix was added in the write.

Source. The report is built from one manuscript, number 38 of batch 7301 of ProveIt’s incoming-reports intake: the archive `OEIS_SquareRoot_Partition_Research.zip` (27 files, a 27-page PDF), which arrived in commit `f8c3a392a` and was placed in the collection by commit `9df4ba51a`. Its pinned repository commit is `1f1981f682b2878bde51a6ad40c22777f362fc05`; the delivered `PROVENANCE.md` calls it a “recursive-tree checkpoint”, but it is a commit (2026-10-01). The delivered author line, kept on the title page, names no assistant; like every manuscript of the intake, the report is AI-assisted and unrefereed.

What the write changed. Every one of the 125 delivered labels received the prefix `srp:`, and every reference was updated with it; this appendix adds one label. The write added the [Write note] paragraphs (after the abstract, after proposition 11.2, in section 13, in section 14, in appendix B and here), three bibliography entries ([11, 12, 13]) and the `\writenote` macro. It also added `cleveref` type hints (`\label[lemma]`, `\label[proposition]`) to the eleven labels of lemmas and propositions, and `\crefalias{section}{appendix}` after `\appendix`: lemmas, propositions and appendices share counters with theorems and sections, and the delivered PDF printed every reference to them as “theorem” or “section”. Only those printed names changed; no label name or number did. No delivered sentence, statement, proof, table or figure was changed or removed.

Delivered names and shipped paths. `article.tex` and `README.md` stay at the report root (the `README` is rewritten as the collection’s report guide); `PROVENANCE.md`, `VALIDATION.md` and `figures/` (two PDF figures, two PNG previews) stay at the root, byte-identical. `verify.py`, `partition_asymptotics.py`, `certificate.py` and `make_figures.py` moved to `code/`; the eleven files of `results/` and `requirements.txt`, `requirements-figures.txt` moved to `data/`, all byte-identical. Not shipped: the delivered `article.pdf` (replaced by a build of this edition) and the checksum ledger `SHA256SUMS` (verified 26/26 at intake, then retired). The delivered `PROVENANCE.md`, `VALIDATION.md`, this article’s section 13 and appendix B and the scripts’ defaults still use the delivery names `results/` and top-level scripts.

Relation to repository material. Two passages of this report have relatives elsewhere in ProveIt. Proposition 11.2 is an instance of the transseries volume’s staircase theorem [11], whose order-theoretic part is formalized [12] (note after proposition 11.2). Proposition 4.1 splits off $h(z) = \log(z/(1 - e^{-z}))$ before an Euler–Maclaurin expansion of a finite q -Pochhammer product; the Fabius tree’s *Gaussian Coefficient Calculus* draft uses the same device for the Gaussian binomial at real argument in its Theorem `q3:thm:double-scaling` [13]. That is a sibling lemma for a different object, not the same theorem.

No repository report treats A097356, A206226, A206227, A206240 or A258268, and none of this report's statements is formalized.

References

- [1] The OEIS Foundation Inc., *A097356: Number of partitions of n into parts not greater than $\sqrt{n}$* . Entry by Reinhard Zumkeller; asymptotic comment by Václav Kotešovec, January 8, 2024. Inspected October 1, 2026. <https://oeis.org/A097356>.
- [2] The OEIS Foundation Inc., *A206226: Number of partitions of n^2 into parts not greater than n* . Entry by Paul D. Hanna; leading asymptotic and general- j formulas by Václav Kotešovec, 2014–2015. Inspected October 1, 2026. <https://oeis.org/A206226>.
- [3] The OEIS Foundation Inc., *A206227: Number of partitions of $n^2 + n$ into parts not greater than n* . Leading asymptotic recorded September 7, 2014. Inspected October 1, 2026. <https://oeis.org/A206227>.
- [4] The OEIS Foundation Inc., *A206240: Number of partitions of $n^2 - n$ into parts not greater than n* . Leading asymptotic recorded September 7, 2014. Inspected October 1, 2026. <https://oeis.org/A206240>.
- [5] The OEIS Foundation Inc., *A258268: Decimal expansion of the constant appearing in the asymptotic growth of A206226*. Inspected October 1, 2026. <https://oeis.org/A258268>.
- [6] E. Rodney Canfield, *From recursions to asymptotics: On Szekeres' formula for the number of partitions*, Electronic Journal of Combinatorics **4**(2) (1997), R6, 16 pp. <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v4i2r6>.
- [7] Dan Romik, *Partitions of n into $t\sqrt{n}$ parts*, European Journal of Combinatorics **26**(1) (2005), 1–17. DOI: 10.1016/j.ejc.2004.02.005. Author's copy: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/pofnk.pdf>.
- [8] Tiefeng Jiang and Ke Wang, *A generalized Hardy–Ramanujan formula for the number of restricted integer partitions*, arXiv:1805.06108. <https://arxiv.org/abs/1805.06108>.
- [9] NIST Digital Library of Mathematical Functions, §24.8, *Series expansions*, Bernoulli Fourier series. Inspected October 1, 2026. <https://dlmf.nist.gov/24.8>.
- [10] Vladimir Reshetnikov, *ProveIt*, repository checkpoint 1f1981f682b2878bde51a6ad40c22777f362fc05, inspected October 1, 2026. Relevant documentation: `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md`. <https://github.com/VladimirReshetnikov/ProveIt/tree/1f1981f682b2878bde51a6ad40c22777f362fc05/Analysis/Transseries>.
- [11] [Added in the write.] ProveIt, *Transseries and Inversion* volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, Theorem `p0:thm:staircase` (staircase recovery and the separation condition).
- [12] [Added in the write.] ProveIt, Lean file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, declarations `Fabius.staircase_ceil` and `Fabius.staircase_separation`.
- [13] [Added in the write.] ProveIt, Fabius-tree draft *Gaussian Coefficient Calculus*, `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Gaussian_Coefficient_Calculus/Gaussian_Coefficient_Calculus.tex`, Theorem `q3:thm:double-scaling`.